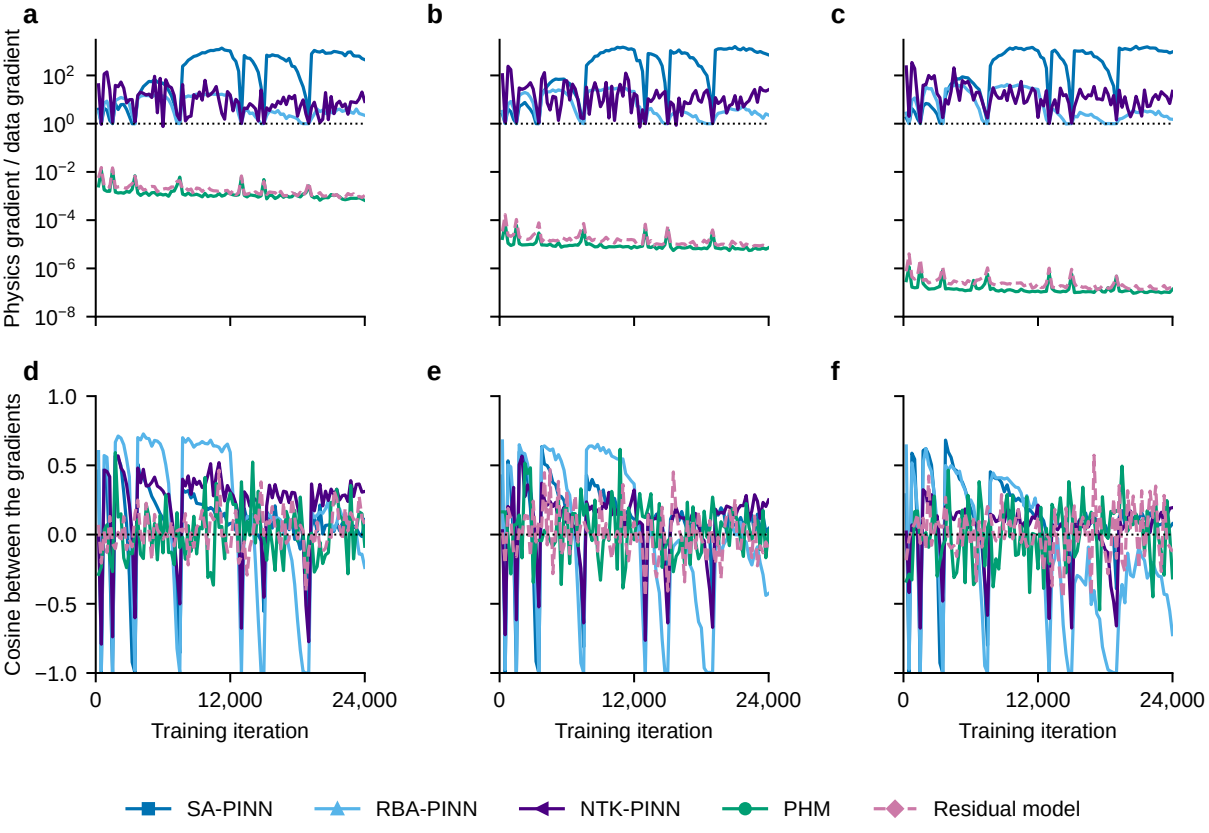
a
b
c
d
e
f
Physics gradient / data gradient
10^{2}
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
Cosine between the gradients
1.0
0.5
0.0
−0.5
−1.0
0
12,000
24,000
Training iteration
SA-PINN
RBA-PINN
NTK-PINN
PHM
Residual model